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Litter Detection Using YOLOv8 on the TACO Dataset Syed Abdullah Student at National University of Computer and Emerging Sciences (NUCES) abdullahsyed2005@gmail.com Abstract—Litter pollution is a growing environmental concern, necessitating automated detection systems for monitoring and cleanup. This project explores object detection using YOLOv8 [1] on the TACO dataset [2], aiming to identify and classify different types of litter in images. Initial experiments used the full TACO dataset (60 classes) with severe class imbalance. Later, a reduced and refined dataset with 18 classes was used to improve class balance. Multiple models were trained and evaluated; results show notable improvements in mAP, precision, and recall after dataset refinement and careful hyperparameter tuning. This report documents dataset preparation, model training methodology, experiments, and conclusions. Index Terms—litter detection, object detection, YOLOv8, TACO dataset, computer vision, dataset balancing I. INTRODUCTION Litter pollution affects ecosystems, cityscapes, waste management systems, and public health. Reliance on manual inspection makes monitoring slow, inconsistent, and resource intensive. Modern computer vision methods—particularly deep learning-based object detectors—offer scalable alternatives that can assist in automated waste detection and environmental maintenance. YOLO (You Only Look Once) models are widely adopted for real-time object detection because they offer strong accuracy with low inference latency. The TACO (Trash Annotations in Context) dataset provides labeled images of many types of waste items and serves as a suitable benchmark for evaluating such models. However, the original TACO dataset contains 60 highly imbalanced classes. Many classes contain very few samples, causing unstable gradients and poor per-class performance in early experiments. To address this, a refined version containing 18 classes with more uniform sample counts was created. This modification produced substantially better detection performance and more reliable training behavior. II. PROPOSED METHODOLOGY The methodology follows a structured approach, including dataset preparation, annotation conversion, model selection, and comparative experimentation. A. Dataset Preparation The original TACO dataset [3] is annotated in COCO format. YOLOv8 requires label files in YOLO format, so annotations were converted accordingly. Each image was paired with a text file containing the class index and normalized bounding-box coordinates. Preprocessing steps included: • Removing corrupted or unreadable images. • Standardizing directory structure for images and labels. • Applying a 70/15/15 train-validation-test split. • Using augmentations such as color jitter, flips, and rotations. Mosaic and mixup were applied selectively. Because severe class imbalance affected learning, an alternative 18-class refined dataset was also used. This improved learning stability and boosted detection accuracy across most classes. Fig. 1: Example class distribution from the TACO dataset [4] B. Model Selection YOLOv8 was selected due to its strong performance and efficient training pipeline. Multiple variants were trained. All models were trained on epochs=100, batch size=16, and image size= 640. All experiments were logged through MLflow and tracked via DagsHub for versioning of models, metrics, and configuration. C. Training Setup Table I summarizes the principal training configurations used. TABLE I: Model training configurations (summary) Name Base Model Epochs

Batch Img LR Optimizer Augmentation yolov8n baseline yolov8s baseline
yolov8s adam optimizer yolov8s adamw tuned yolov8s heavy augmentation
yolov8n.pt 100 yolov8s.pt 100 yolov8s.pt 100 yolov8s.pt 100 yolov8s.pt 100
16 640 16 640 16 640 16 640 16 640 0.01 SGD 0.01 SGD 0.001 Adam 0
.0005 AdamW 0.01 SGD None None None Heavy Heavy Heavy augmentation
included mosaic and mixup, improving robustness to occlusion and clutter. III.
EXPERIMENTS AND RESULTS We split the experimental narrative into two
phases: Initial experiments on the full 60-class TACO dataset, and Refined
experiments on the reduced 18-class dataset. A. Initial Experiments (60
classes) Trained models on the original TACO (60 classes). Severe class
imbalance impacted per-class AP. The first phase of experimentation used the
complete 60- class TACO dataset. The extreme imbalance across categories
had a substantial impact on detection performance, particularly for classes
with very few samples. • Classes with minimal representation showed near-
zero average precision; for example, the Battery category contained only two
annotated instances, making reliable learning impossible. • YOLOv8s
consistently outperformed the YOLOv8n base- line in overall precision and
mAP, although its validation loss occasionally fluctuated due to the unstable
class distribution. • Models trained with the Adam optimizer achieved notice-
ably higher recall but tended to trade off some precision compared to SGD-
based runs. • Data augmentation, including mosaic, flips, and color jitter,
enhanced generalization and mitigated some im- balance effects when applied
conservatively. Figure 2 shows a schematic comparison of metrics across the
5 models. Fig. 2: Training and validation metrics (precision, recall, mAP,
losses) across initial experiments. Representative numeric results for the 60-
class experiments: TABLE II: Representative results initial experiments (60
classes) Model Precision Recall mAP50 yolov8n baseline yolov8s baseline
yolov8s adam optimizer yolov8s adamw tuned yolov8s heavy augmentation
0.438 0.123 0.449 0.151 0.359 0.176 0.474 0.144 0.296 0.187 0.115 0.138
0.162 0.174 0.151 Analysis: the nano model (yolov8n) had the highest
precision but low recall, it missed many litter instances. The yolov8s variants
provided a better balance for deployment. B. Refined Dataset Experiments
(18 classes) To reduce imbalance, the dataset was reduced to 18 classes
using a YOLO-formatted version [5]. Models were retrained with similar
hyperparameters and assessed. • Class distribution was more even, allowing
models to learn underrepresented classes. • Overall mAP, recall, and per-class
AP increased across the board. Fig. 3: Class distribution (18-class refined
dataset) [6]. Figure 4 shows a schematic comparison of metrics across the 4
models. Fig. 4: Training and validation metrics (precision, recall, mAP, losses)
across initial experiments. TABLE III: Performance metrics for refined 18-
class models Model mAP50 Precision Recall yolov8s adam optimizer mod
yolov8s baseline mod yolov8n baseline mod yolov8s modified8 0.411 0.663
0.423 0.650 0.369 0.683 0.491 0.633 0.387 0.394 0.339 0.400
Observations: • The yolov8s modified8 model (heavy augmentation) achieved
the highest mAP50 (0.49) and recall (0.40), making it the recommended
deployment candidate. • The nano model remained a precision specialist but
had low recall (misses too many objects). • Heavy augmentations increased
training loss in some cases but reduced validation classification loss,
indicating better generalization. C. Qualitative Results Sample detection
outputs from the best model are shown in Figure. (a) Detection sample 1 (b)
Detection sample 2 Fig. 5: Detections by the deployed model (yolov8s
modified8). D. Additional Diagnostic Figures To further examine model
behavior and parameters, all experiment logs, metrics, and models are
available through MLflow on the DagsHub project repository [7]. IV.
CONCLUSIONS AND FUTURE WORK • Reducing class count to 18 and
improving class balance produced significant gains in detection metrics. •
Heavy data augmentations (mosaic, flips, rotations) com- bined with YOLOv8s

gave the best operational balance of recall and precision. • Future work: collect more labeled images for rare classes, explore class-aware sampling and loss weighting, and integrate detectors into mobile or robotic cleanup systems. ACKNOWLEDGMENTS Experiment logs and models were tracked using MLflow/DagsHub. Thanks to dataset authors and maintainers of the YOLO ecosystem. REFERENCES [1] Ultralytics, "Ultralytics yolov8 documentation," <https://docs.ultralytics.com/>, 2023, accessed: 2025-12-10. [2] L. Bossard, M. Guillaumin, and L. V. Gool, "Taco: Trash annotations in context," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (CVPR) Workshops, 2019, dataset: Trash Annotations in Context (TACO). [3] kneroma, "Taco trash dataset," [Kaggle dataset](https://www.kaggle.com/datasets/kneroma/tacotrashdataset), <https://www.kaggle.com/datasets/kneroma/tacotrashdataset>, 2025, accessed: 2025-12-10. [4] A. Syed, "Taco dataset processing notebook," <https://www.kaggle.com/code/syedabdullah1247/official-taco-dataset>, 2025, accessed: 2025-12-10. [5] V. Lanz, "Taco dataset (yolo format)," [Kaggle dataset](https://www.kaggle.com/datasets/vencerlanz09/taco-dataset-yolo-format), <https://www.kaggle.com/datasets/vencerlanz09/taco-dataset-yolo-format>, 2023, accessed: 2025-12-10. [6] nujhatraisaarpa, "Taco trash-detection yolo12n notebook," [Kaggle notebook](https://www.kaggle.com/code/nujhatraisaarpa/taco-trash-detection-yolo12n), <https://www.kaggle.com/code/nujhatraisaarpa/taco-trash-detection-yolo12n>, 2025, accessed: 2025-12-10. [7] A. Syed, "Litter detection mlflow runs (project logs)," <https://dagshub.com/abdullahsyed2005/litter-detection.mlflow/#/experiments/7>, accessed: 2025-12-10.